PHYSICS

SECTION – I
Single Correct Choice Type

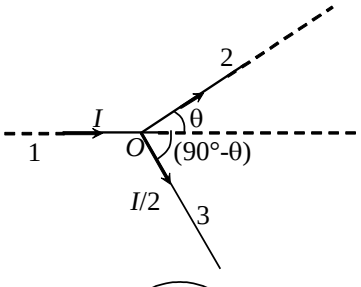
This section contains 8 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

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1. Find the value of θ in the range of $0 \le \theta \le \frac{\pi}{2}$, so that the magnetic field is maximum at point *P* (0, 0, *d*)

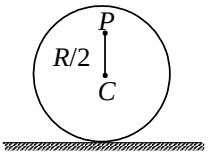
(a) 45° (b) 60°

(c) 30° (d) 90°



2. A cylinder of mass *M* and radius *R* is rolling over a smooth horizontal surface. A point *P* is on the cylinder a distance $\frac{R}{2}$ above the center of the cylinder. The angular momentum of the cylinder about point *P* is (the velocity of its CM is v_0)

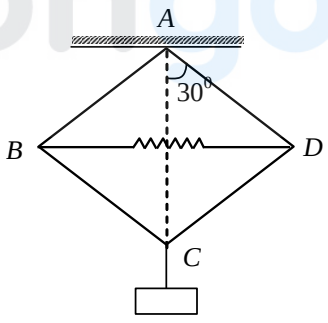
(a) zero (b) $\frac{MRv_0}{4}$ (c) MRv_0 (d) $\frac{3}{4}mRv_0$



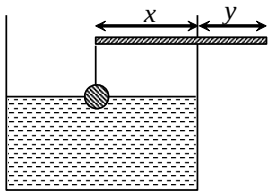
3. Four weightless rods of length *l* each are connected by hinged joints and forms a rhombus. Hinge *A* is fixed and load is suspended from hinge *C*. Hinge D and B are connected by weightless spring of length 1.5 *l* in unstretched state. In equilibrium the rod forms an angle 30^0 with vertical, then the time period of oscillation of load is

(a) $T = 2\pi\sqrt{\frac{l\sqrt{3}}{g}}$ (b) $T = 2\pi\sqrt{\frac{3\sqrt{3}l}{2g}}$

(c) $T = 2\pi\sqrt{\frac{l}{2\sqrt{3}g}}$ (d) $T = 2\pi\sqrt{\frac{l\sqrt{3}}{5g}}$



4. A homogenous aluminium ball of radius r = 0.5cm is suspended by a weightless thread from an end of a homogenous rod of mass M = 4.4 gm. The rod is placed on the edge of a tumbler with water so that half of the ball is submerged in water when the system is in equilibrium (figure shown). Densities of aluminium and water is 2.7 × 10^3 kg /m^3 and 10^3 kg/m^3 respectively. The ratio $\frac{y}{x}$ of the parts of the brim is approximately



(a) $\frac{3}{2}$ (b) $\frac{2}{3}$ (c) $\frac{10}{27}$ (d) $\frac{27}{10}$

5. Temperature dependence of speed of sound could be expressed as $v = k\sqrt{T}$, where k is some positive constant. In a medium sound is traveling along x-axis. At x=0 temperature is T_1 and at $x = l$, temperature is T_2 and the temperature varies linearly with x. Time taken by sound to travel from $x = 0$ to $x = l$ is

(a) $\frac{2l}{k\left(\sqrt{T_1}+\sqrt{T_2}\right)}$ (b) $\frac{l}{k\left(\sqrt{T_1}+\sqrt{T_2}\right)}$ (c) $\frac{l}{2k\left(\sqrt{T_1}+\sqrt{T_2}\right)}$ (d) $\frac{1}{2k\left(\sqrt{T_1}+\sqrt{T_2}\right)}$

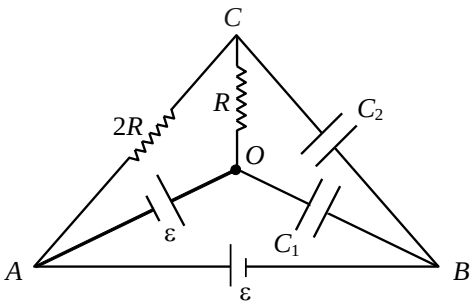
6. Molar heat capacity of gas whose molar heat capacity at constant volume is C_V for process $P = 2e^{2V}$ is

(a) $C_V + \frac{R}{1+2V}$ (b) $C_V + \frac{R}{2V}$ (c) $C_V + \frac{R}{V}$ (d) none of these

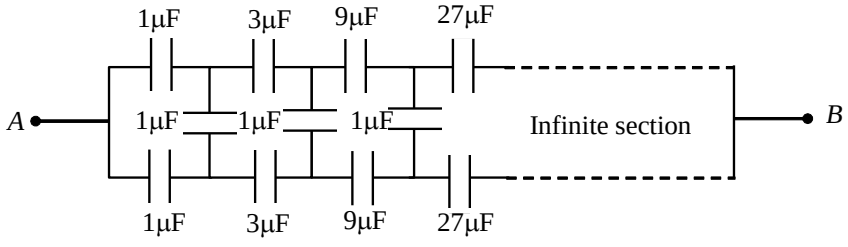
7. If energy stored in the capacitor C_1 and C_2 are same then what is the value of C_1 / C_2 ?

(a) $\frac{25}{36}$ (b) $\frac{1}{36}$

(c) $\frac{4}{9}$ (d) $\frac{9}{4}$



8. What is the equivalent capacitance between the points *A* and *B*?



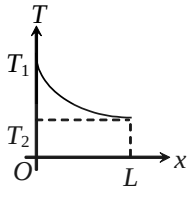
(a) $\frac{2}{3}\mu F$ (b) $\frac{4}{3}\mu F$ (c) infinite (d) $\left(1+\sqrt{3}\right)\mu F$

SECTION – II
Multiple Correct Choice Type

This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

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9. The two ends of a long bar maintained at different temperature. The graph of temperature against distance of the bar when it has attained steady state is shown in the figure. Then select the correct alternative(s)



(a) the cross-section area of the bar may increase as the distance from the hot end increases
(b) the cross-section area of the bar may decreases as the distance from the hot end increases
(c) the conductivity of the rod may be increasing as the distance from hot end increases
(d) the conductivity of the bar may decrease as the distance from hot end increases

10. Two flat worms *A* and *B* each of mass 20 grams climb over a very thin wall of height 10 cm. The worm *A* is 20 cm long and the worm *B* is wider and only 10 cm long. Work done by gravity on the worm *A* is W_1 when half of its is over the top of the wall and W_2 be work done by the worm *B* when half of it is over the top of the wall, then ($g = 10\ m/s^2$)

(a) $W_1 = -10.0$ mJ (b) $W_2 = -15.0$ mJ
(c) $W_2 = 15.0$ mJ (d) $W_1 = 20.0$ mJ

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11. The threshold wavelength for photoelectric emission from a material is 6000 A°. Photo electrons will be emitted when this material is illuminated with a laser beam of
(a) red colour and having power 50 W
(b) blue colour and having power 50 W
(c) blue colour and having power 0.005 W
(d) violet colour and having power 5 W

12. A diatomic gas goes through a process $\Delta U + \lambda W = 0$. Now if
(a) $\lambda = 1$, then process is adiabatic
(b) $\lambda = 0$, then process is isochoric
(c) $\lambda = -\frac{5}{2}$, then the process is isobaric
(d) $\lambda = \frac{5}{2}$, then the process follows $PV^2 =$ constant

SECTION–III
Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

Passage-I

A particle *P*, of charge *q* and mass *m*, is placed at a point on a smooth horizontal surface and is free to move. Another particle *Q*, of same charge and mass 2*m* is projected from a distance *r* from *P* with an initial speed v_0 directly towards *P*. As *Q* approach *P*, first the separation between them decreases and after that increases.

13. What is the speed of particle *P* when separation between them is minimum?
(a) zero (b) $\frac{v_0}{2}$ (c) $\frac{v_0}{4}$ (d) $\frac{2v_0}{3}$

14. At the closest separation, the potential energy of the system of particles is (Here r_0 is minimum separation between the particles)

(a) $\frac{q^2}{4\pi \in_0 r}$ (b) $\frac{q^2}{4\pi \in_0 r}+\frac{mv_0^2}{3}$

(c) $\frac{-q^2}{4\pi \in_0 r_0}+\frac{1}{4}mv_0^2$ (d) $\frac{q^2}{4\pi \in_0 r}+\frac{1}{2}mv_0^2$

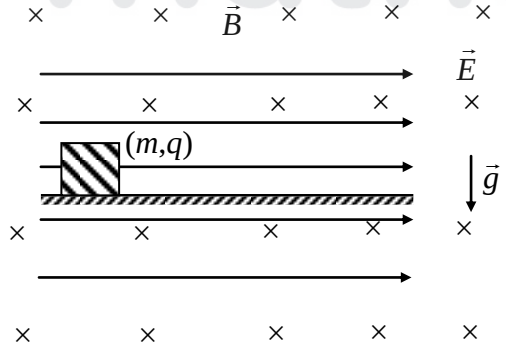
15. Separation between the two particles when speed of P is $\frac{v_0}{2}$

(a) $\frac{q^2}{\frac{q^2}{r}+\frac{5mv_0^2\pi \in_0}{4}}$ (b) $\frac{1}{\frac{1}{r}+\frac{3mv_0^2\pi \in_0}{4q^2}}$

(c) $\frac{16qr}{4q^2+5mv_0^2\pi \in_0 r}$ (d) $\frac{r}{10}$

Passage-II

A small insulating block having mass m = 100 gram carrying charge q = +1mC is placed at rest on a smooth insulating horizontal surface in uniform vertical gravitational field (g = 10 ms^{-2}). At t = 0, uniform and constant electric and magnetic field having magnitudes $E = 10^4$ N/C and B = 0.1 Tesla is switched on in the directions as shown and the block is free to move. For subsequent motion of the block in the given situation answer following questions.



16. The block will lose contact with the horizontal surface at

(a) $t = 1$ s (b) $t = 10$ s (c) $t = 100$ s (d) $t = 1000$ s

17. The block will lose contact with the surface after travelling through

(a) 5m (b) 5×10^2m (c) 5×10^4m (d) 5×10^5m

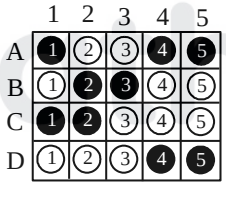
18. Magnitude of time rate of change of normal reaction on the block by the surface will be

(a) 10^{-2} N/s (b) 5×10^{-3} N/s (c) 10^{-3} N/s (d) 0.1 N/s

SECTION–IV
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



1. Column–I contains equation of a particle representing different motions. Here $\vec{v}$ represents instantaneous velocity x is position along x axis, y is position along y axis and t is an instant of time. Column–II contains various characteristics of the motion. Match appropriately and mark your answer

Column–I		Column–II	
(A)	$\vec{v} = 5t\,\hat{i} + 4\,t\,\hat{j}$	**1.**	Straight line motion
(B)	$\vec{v} = -5t\,\hat{i} + 5\,\hat{j}$	**2.**	Motion on parabolic path
(C)	$y = 2x - 5x^2$	**3.**	Acceleration is constant
(D)	$x = 6\sin\frac{2\pi t}{5},\ y = 6\cos\frac{2\pi t}{5}$	**4.**	Velocity is not constant
		5.	Speed is constant

2. When we write expression for energy of electron in n^{th} orbit of helium ion (He^+) we take zero potential energy for $n = \infty$, but the potential energy depends on reference. If we take total energy of atom for $n = 1$ orbit as zero then

Column–I		Column–II	
(A)	Total energy of electron in $n = 2$	**1.**	54.4 eV
(B)	Ionization energy from ground state	**2.**	40.8 eV
(C)	Energy required to excite electron from $n = 1$ to $n = 2$	**3.**	depends on reference level
(D)	Negative of potential energy of electron in $n = 1$	**4.**	independent of reference level.
		5.	65.28×10^{-20} J